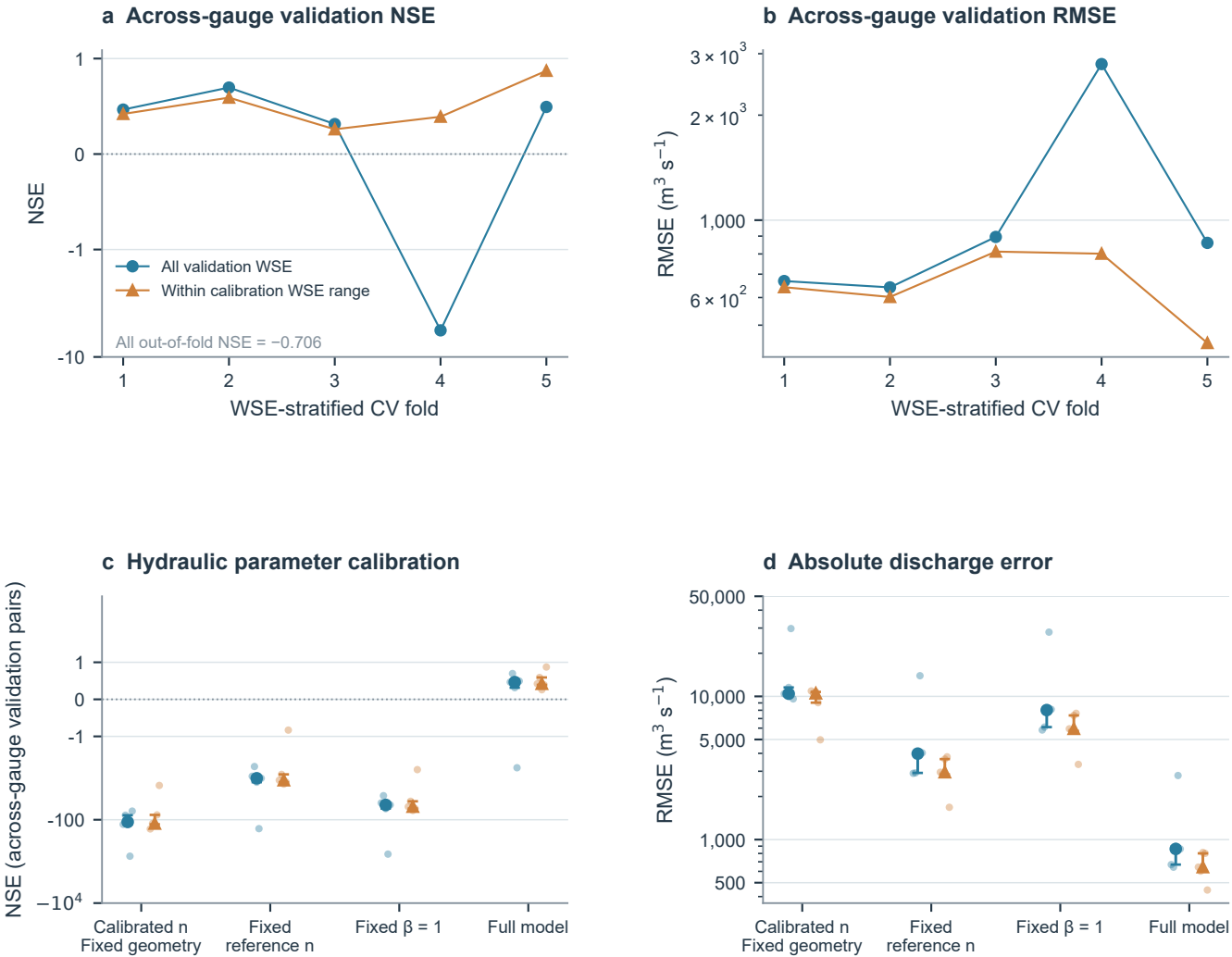

a Across-gauge validation NSE
NSE
1
0
-1
-10
All validation WSE
Within calibration WSE range
All out-of-fold NSE = −0.706
1
2
3
4
5
WSE-stratified CV fold
b Across-gauge validation RMSE
RMSE (m³ s⁻¹)
3 × 10³
2 × 10³
1,000
6 × 10²
1
2
3
4
5
WSE-stratified CV fold
c Hydraulic parameter calibration
NSE (across-gauge validation pairs)
1
0
-1
-100
−10⁴
Calibrated n
Fixed geometry
Fixed
reference n
Fixed β = 1
Full model
d Absolute discharge error
RMSE (m³ s⁻¹)
50,000
10,000
5,000
1,000
500
Calibrated n
Fixed geometry
Fixed
reference n
Fixed β = 1
Full model